

MAT 115E Introduction to Programming Language

Lab-3/ CRN : 23492

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1 Question 1

In the triangle shown, $a = 5$ cm, $b = 7$ cm, $\gamma = 55^\circ$. Define a , b , γ as variables, and then:

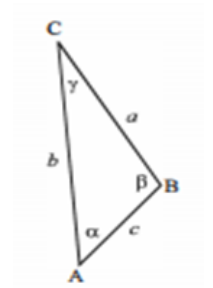
- a.) Calculate the length of c by substituting the variables in the Law of Cosines.

$$\text{Law of Cosines : } c^2 = a^2 + b^2 - 2ab \cos \gamma$$

- b.) Calculate the angles α and β (in degrees) using the Law of Sines.

$$\text{Law of Sines : } \frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

- c.) Check that angle β is between $\frac{\pi}{6}$ and $\frac{\pi}{3}$. Display 1 on the screen if the condition is true; otherwise, display 0.



2 Question 2

Create a C program that accomplishes the followings:

- Read three letters in lexicographical order using at least one **scanf** function and one **getchar** function.
- Determine whether harmonic mean of the first and third character is greater than second character and whether ASCII value of the second character is greater than 100.

$$\text{Harmonic Mean} := hm(x_1, x_2) = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}}$$

- Display 1 on the screen if the condition is true; otherwise, display 0.

Sample Output:

> Enter the first character: f > Enter the second character: g > Enter the third character: h > Is $hm(f, h) > g$ and ASCII of g greater than 100 : 0	> Enter the first character: f > Enter the second character: g > Enter the third character: i > Is $hm(f, i) > g$ and ASCII of g greater than 100 : 1
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3 Question 3

The four-sided geometry shown in Figure consists of two triangles having a common side a . The law of cosines for the top triangle states that

$$a^2 = b_1^2 + c_1^2 - 2b_1c_1 \cos A_1 \quad (1)$$

and a similar equation can be written for the bottom triangle. Develop a procedure for computing the length of side c_2 if you are given the lengths of sides b_1, b_2, c_1 and the angles A_1 and A_2 . Write a C program to implement this procedure using the following values: $b_1 = 180$ m, $b_2 = 165$ m, $c_1 = 115$ m, $A_1 = 120^\circ$, and $A_2 = 100^\circ$.

- a.) Find the length of side a using the law of cosines.
b.) Find the angle C_2 using the sine theorem and the property of the triangle.

